Sensors Administration User Guide (AQDR API Sensors)

This document is intended to serve as a ***User Guide for the administration of the sensor devices*** (*InBiot* and *Mann+Hummel* sensors) for continuous monitoring in the context of the K-HEALTHinAIR project.

These sensors-related operations (e.g., register a sensor, delete a sensor, etc.) must be carried out through the K-HEALTHinAIR Air Quality Data Repository API, whose User Interface (UI) is reachable at:

Link from AQDR UI HOME, or aqdr-api.k-healthinair.ari-health.eu

For now, the interface provided corresponds to the well-known "swagger" documentation and testing tool for APIs based on the *OpenAPI* standard, which provides a convenient mechanism for our purpose of delegating the maintenance of sensor details to those responsible of them.

1. *Authorization and authentication.*
2. *Sensor's operations.*
3. *How to register a new sensor.*
4. *How to delete an already registered sensor.*
5. *Extract information about the registered sensors.*
6. *Pilots' administrators.*

Note that the Air Quality Data Repository API includes several groupings of endpoints but the operations for managing sensors are only and just grouped under the "Sensors" one (*Figure* 1, blue circle). Expanding the section, the available operations (or endpoints) will be listed and documented in the way them could be tested/executed, as will be described later in this document.

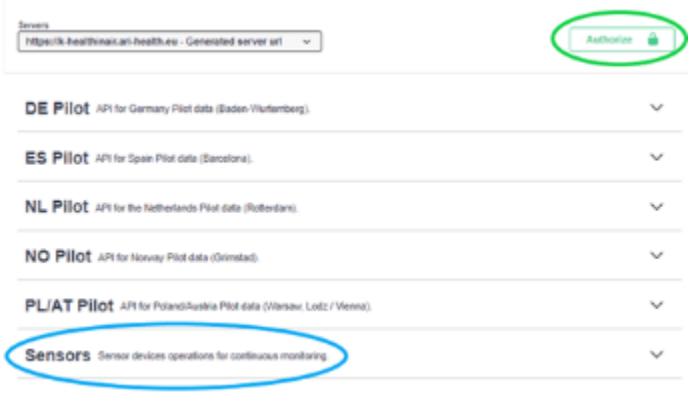


Figure 1: Sensors tab in K-HEALTHinAIR AQDR API User Interface (UI).

1. Authorization and authentication

The *Air Quality Data Repository API* is secured based on authentication scheme which means that all endpoints and operations are protected, and therefore valid credentials are needed to be authenticated. Each one of the assigned responsible for this task has been assigned a particular user and a unique associated password that grants the user enough permissions to perform the delegated operations.

To authenticate, clicking on the padlock icon is needed, either on the click on the top level Authorize button (*Figure 1*, green circle) or on the padlocks of each specific operation (*Figure* 2), and then fill the "username" and "password" fields with the proper credentials and click on Authorize (*Figure* 3).

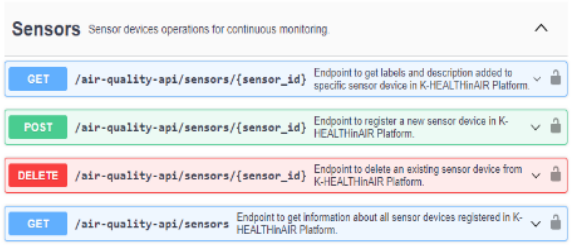


Figure 2: Available operations for sensors list maintenance.

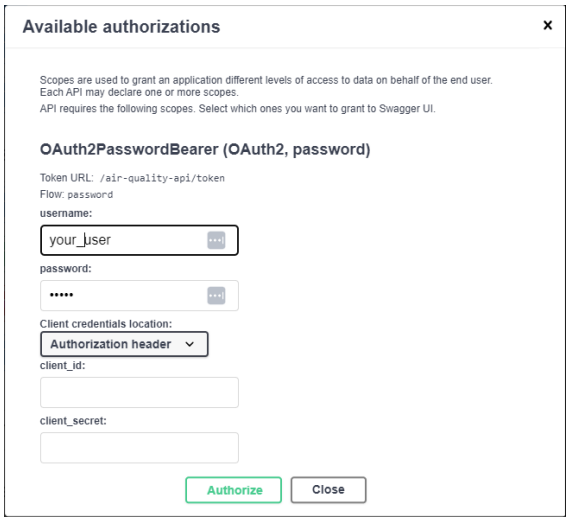


Figure 3: Authentication pop-up to fill in credentials.

Once you are authenticated, you can start performing operations during the following 30 minutes when, for security reasons, the session will expire. After that default time, you may receive an authorization error and, in that case, would be necessary to authenticate again by re-entering the credentials as previously indicated.

2. Sensor's operations

The basic operations for the sensor devices list management are listed under the grouping labelled *Sensors* (*Figure* 2), where the authenticated users can:

- **POST** Register a new sensor device in the K-HEALTHinAIR Platform.
- **DELETE** Remove an already registered sensor device.
- **GET** Retrieve the description and location tags of one or all the registered sensors.

To perform an operation, just click on the specific endpoint, click on the Try it Out button in the right side, complete the required fields (if needed) and click the blue button **Execute** at the end of the section. After every operation performed, either a successful response or an error is returned.

Note that, without switching to the execution mode, what is shown is just documentation of the potential results expected by the endpoint, not to be confused with the result of an actual execution.

3. How to register a new sensor

Registering a new sensor in the platform is performed through the **POST** `/air-quality-api/sensors/{sensor_id}` endpoint. To complete a successful registration, you should fill in all the fields of the form so that the description of the device you want to register is complete. As an example, the *Figure* 4 shows, IB (InBiot) is selected as provider (instead of MH, Mann+Hummel, which is the other alternative) and AT (Austria) as pilot in the dropdowns.

Sensor devices registration must be K-HEALTHinAIR ID Convention-compliant, which means that some fields have automated validation mechanisms that will trigger you errors if non-expected format is provided (e.g., an ID and apiKey that do not match, or a code for a scenario that is not considered in the project). A table with basic generic codes agreed in the project is provided in the API endpoint description (*Figure* 5) and all the fields include an explanation in the API. Nevertheless, "scenario","area" and "inclusion" fields must be completed carefully since there is no actual way to check for the validity and correctness of these labels.

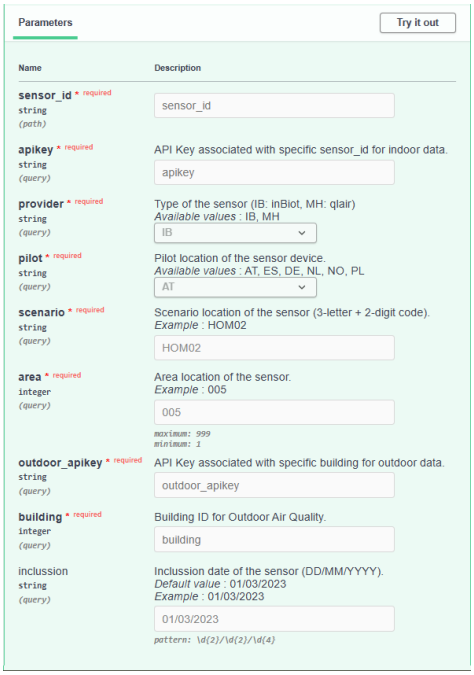
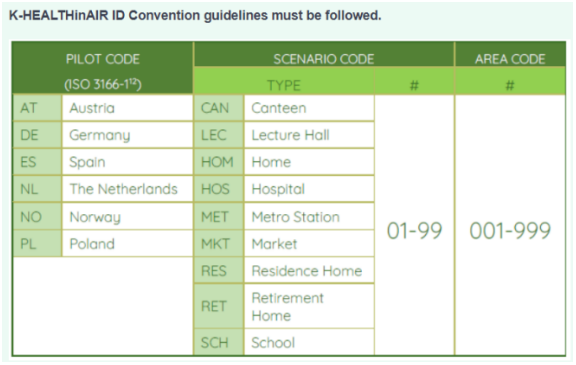


Figure 4: Form fields to be completed in device registration.



PILOT CODE (ISO 3166-1[2])		SCENARIO CODE TYPE		#	AREA CODE #
AT	Austria	CAN	Canteen	01-99	001-999
DE	Germany	LEC	Lecture Hall		
ES	Spain	HOM	Home		
NL	The Netherlands	HOS	Hospital		
NO	Norway	MET	Metro Station		
PL	Poland	MKT	Market		
		RES	Residence Home		
		RET	Retirement Home		
		SCH	School		

Figure 5: Sensor device ID Convention.

The registration process must be performed device by device. This is designed in this way to offer individual validation mechanisms, to avoid errors in registration of several sensors at the same time, and to avoid accidentally affecting other already registered sensors.

Please notice that once you successfully register a device (i.e., Code 200 as confirmation), the values of the form fields remain the same. This may be useful if you are registering a set of sensors of the same location, and thus it could be enough to change only several fields (e.g., the sensor device identifier and its apiKey only) for registering the next sensor in your list (i.e., other fields like pilot or scenario could be the same, so typing these fields again could be omitted). For this reason, registering sensors in a coherent order could be helpful for you to save time and perform this process in a more efficient manner.

It is not possible to register a sensor without filling in all the fields, since they are all mandatory. You may receive an error response if you try to register incorrect information.

A basic verification of the existence of that sensor_id against the provider's API will be carried out to validate it, rejecting it if this is not possible with the error 400 – Invalid sensor id. Also, when trying to register a sensor device that is already registered, an error code will be returned (`Code 400 Bad Request: "This sensor_id is already registered"`). To update a sensor device, and due to its potential consequences, refer to the deletion of sensor instructions below.

4. How to delete an already registered sensor

Deleting a new sensor in the platform is performed through the **DELETE** `/air-quality-api/sensors/{sensor_id}` endpoint. To complete a successful deletion, the specific identifier (sensor_id) of the device is the only necessary field to be completed in the form (*Figure* 6).

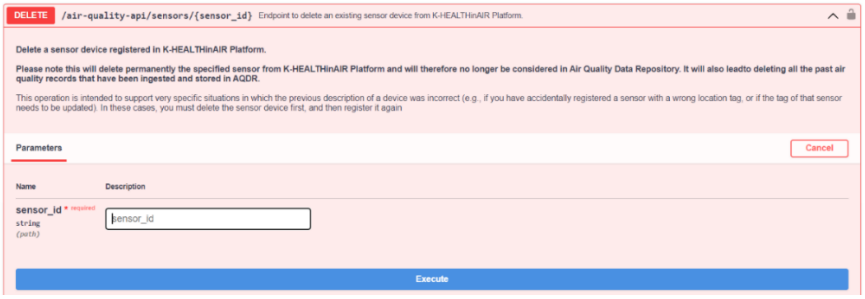


Figure 6: Endpoint for sensor deletion from K-HEALTHinAIR Platform.

This operation is intended to support very specific situations in which the previous description of a device was incorrect (e.g., if you have accidentally registered a sensor with a wrong location tag/label, or if the tag of that sensor needs to be updated). In these cases, you must delete the sensor device first, and then register it again using the form in the POST endpoint as previously explained. Note that deleting a sensor device will cause not only to unregister it from the registry list of the sensors, but also to delete all the related past air quality records that have been ingested. Nevertheless, this would not be a major issue, since if a sensor device is accidentally deleted, it could be registered again as explained before with its new description due to INB and M+H sensors store every historic record in the device.

If you try to delete a sensor device that is not registered in the K-HEALTHinAIR Platform, you will receive an error (`Code 404 Not Found`).

5. Extract information about the registered sensors

Retrieving the information of a specific sensor device or of all the sensor devices that are currently registered in the platform is possible through the **GET** endpoints and this could be very useful to check the correctness and validity of the information of the sensors that have been already registered and ensuring the information has been correctly ingested. There are two possibilities:

- The **GET** `/air-quality-api/sensors` endpoint for all the sensor devices that are currently registered in the platform. No input parameters are needed, and an Excel file will be provided as a result ready for download in which sensor devices information can be examined.
- The **GET** `/air-quality-api/sensors/{sensor_id}` endpoint to retrieve the information of just a single specific sensor device, providing as parameter its unique identifier. In the same way as in the previous case, the file will be available for download and examination.

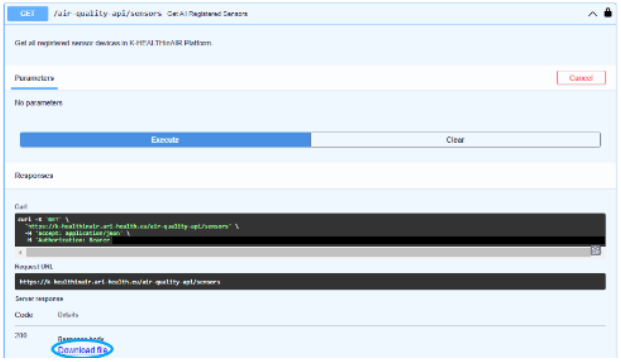


Figure 7: Method for retrieve all sensors description.

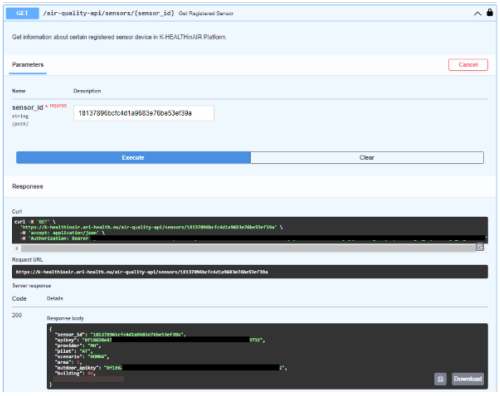


Figure 8: Method for retrieve one sensor description.

6. Pilots' administrators

With access to its associated pilot only:

- AT: *alexandra.kristian@meduniwien.ac.at*
- DE: *johannes.dalheimer@i-2-m.eu*
- ES: *snikolaisen@inbiot.es*
- NL: *s.c.deleede@erasmusmc.nl*
- NO: *tore.vehus@uia.no*
- PL: *mariusz.rogulski@pw.edu.pl* and *anna.pisarska@imp.lodz.pl*

Besides, users for *InBiot* and *Mann+Hummel* sensor providers have been also created (with access to their associated pilots only):

- INB: *mfigols@inbiot.es* and *snikolaisen@inbiot.es*
- M+H: *johannes.dalheimer@i-2-m.eu*

This document is attached and accessible from the *Sensors > >POST* section description.